Use Cases

for

<URECAMatch>

Version 1.0 approved

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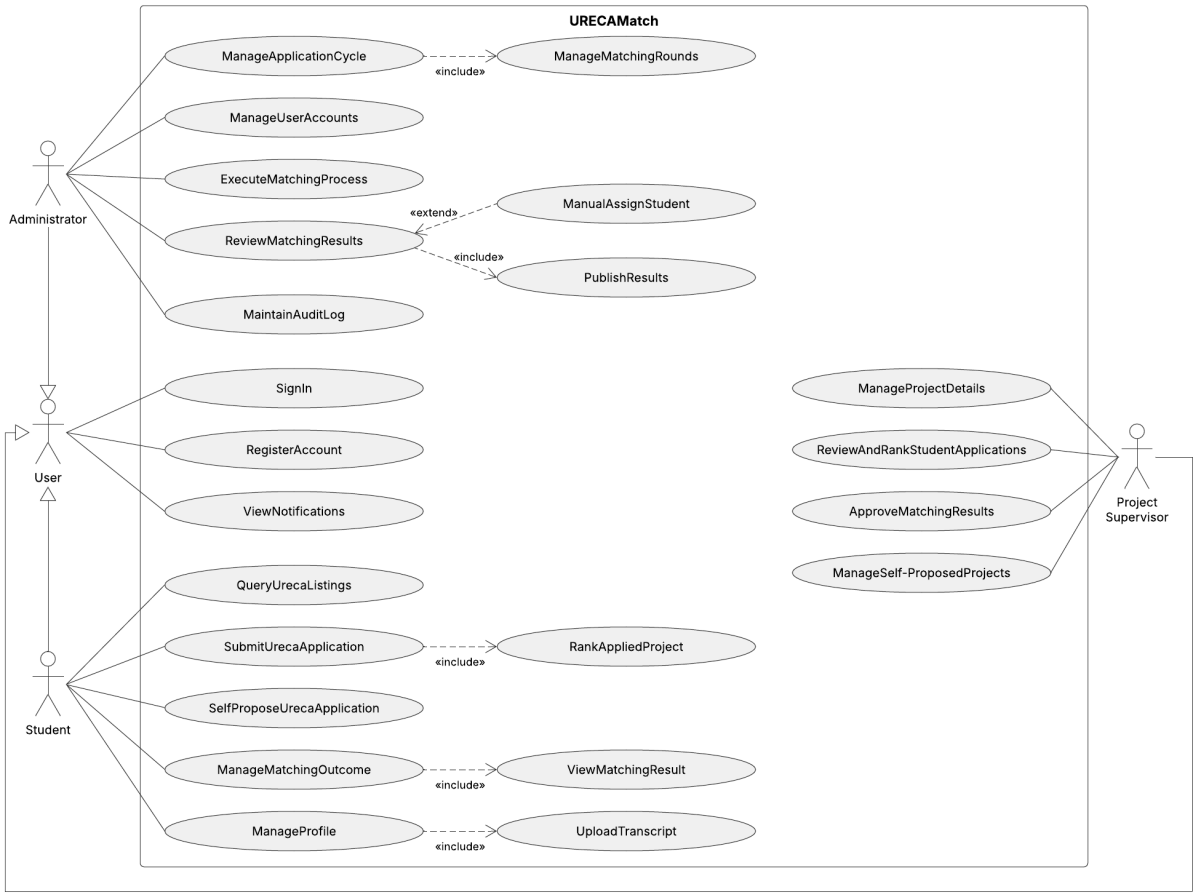
SCEF GRP 4

<2/9/2026>

Revision History

Name	Date	Reason For Changes	Version
Ian Tan	19/10/2026	LAB 3: use cases aligned with SRS Section 4	1.1

1. Use Case Diagram



2. Use Case Description

2.1 For Functional Requirement #1 (Admin)

2.1.1 ManageApplicationCycle

Use Case ID:	#1-1
Use Case Name:	ManageApplicationCycle

Created By:	Nicholas Ng	**Last Updated By:**	Ian Tan
Date Created:	26th August 2026	**Date Last Updated:**	19th October 2026

Actor:	Admin
Description:	Admin creates a new Application Cycle or modifies an existing one. An Application Cycle consists of a global start/end date, a pre-application viewing period (where project listings are visible but applications are disabled), and up to three sequential matching rounds (each comprising an Application Window followed by a Review Window, with lengths set by the Admin).
Preconditions:	The Admin is signed in.
Postconditions:	The overall Application Cycle and its constituent rounds are successfully created or updated in URECAMatch
Priority:	High
Frequency of Use:	Several times per year.
Flow of Events:	1. The Admin navigates to the Application Cycles tab and selects either "New Application Cycle" or an existing cycle to modify. 2. The System displays the Application Cycle configuration page. 3. The Admin configures the global cycle parameters, including the overall start date, end date, and the specific dates for the pre-application viewing period. 4. The Admin configures the individual matching rounds (ManageMatchingRounds). 5. The Admin clicks "Create" or "Save" to submit the configuration. 6. The System validates the provided parameters and displays a confirmation prompt. 7. The Admin clicks "Confirm" to proceed. 8. The System saves the Application Cycle and updates the active records.
Alternative Flows:	AF-1: The Admin clicks “Cancel”

	1. The System confirms the cancellation via a prompt. 2. The Admin acknowledges the cancellation. 3. The System discards filled fields or changes and exits to the main page.
Exceptions:	E1: The closing date of an Application Cycle is earlier than the start date. 1. The System rejects this change. 2. The System displays the rejection reason to the Admin, and remains on the configuration page. E2. The Admin attempts to create a new Application Cycle with dates that overlap an existing Application Cycle. 1. The System rejects this change. 2. The System displays the rejection reason to the Admin, and remains on the configuration page. E3. The Admin attempts to create a new Application Cycle with a closing date that has already lapsed 1. The System rejects this change. 2. The System displays the rejection reason to the Admin, and remains on the configuration page. E4: The Admin made no changes to the Application Cycle. 1. The System greys out the “Save” button and disables its interactivity.
Includes:	ManageMatchingRounds
Special Requirements:	If the closing date of an Application Cycle is modified during its active period, it shall prompt the Admin to send out a notification to affected Users.
Assumptions:	Only one Application Cycle shall be active at any given time.
Notes and Issues:	-

2.1.2 ManageMatchingRounds

Use Case ID:	#1-2		
Use Case Name:	ManageMatchingRounds		
Created By:	Edwin	**Last Updated By:**	Ian Tan

Date Created:	6th September 2026	**Date Last Updated:**	19th October 2026

Actor:	Admin
Description:	Admin configures up to three sequential matching rounds within the active application cycle. Each round consists of an Application Window (1 to 14 days) followed by a Review Window (1 to 7 days), with lengths set by the Admin.
Preconditions:	The Admin is currently executing the ManageApplicationCycle use case.
Postconditions:	All matching rounds are valid and match to the whole application cycle.
Priority:	High
Frequency of Use:	Several times per year.
Flow of Events:	1. The Admin selects the option to "Add/Edit Matching Rounds" within the cycle configuration interface. 2. The System generates the interface for up to three sequential matching rounds. 3. For each round, the Admin sets the Application Window length (1 to 14 days), the Review Window length (1 to 7 days) and, optionally, a start date. 4. The Admin chooses whether the applications of unmatched eligible Students roll over into the next round. 5. The System calculates the window dates of each round from its start date and window lengths. 6. The Admin clicks to confirm the round configuration. 7. The System validates that none of the rounds overlap with each other and that all rounds fall within the global Application Cycle dates.
Alternative Flows:	AF-1: Admin configures fewer than three rounds

	1. The Admin leaves the start date for Round 2 or Round 3 blank. 2. The System configures only the populated rounds for the cycle.
Exceptions:	E1: The calculated dates for a matching round overlap with another matching round. 1. The System rejects the configuration and highlights the overlapping dates. E2: The matching round dates fall outside the global Application Cycle start and end dates. 1. The System rejects the configuration and prompts the Admin to adjust the round dates or the global cycle dates.
Includes:	-
Special Requirements:	A Matching Round can be changed or removed only before its Application Window starts. If the rounds are modified during an active cycle, the System shall prompt the Admin to notify affected Users.
Assumptions:	Every matching round follows the same rules, but each may have its own window lengths and start date.
Notes and Issues:	-

2.1.3 ExecuteMatchingProcess

Use Case ID:	#1-3		
Use Case Name:	ExecuteMatchingProcess		
Created By:	Edwin Tan	**Last Updated By:**	Ian Tan
Date Created:	29th August 2026	**Date Last Updated:**	19th October 2026

Actor:	Admin
Description:	Admin runs the stable-matching algorithm to assign students to projects based on ranked preferences and supervisor rankings
Preconditions:	1. The Admin is signed in. 2. The current matching round's Review Window has ended.
Postconditions:	A proposed allocation is generated where no project exceeds available capacity and each student receives at most one match. Open slots carry over to the next round, and if the Admin enabled rollover, the applications of unmatched eligible Students carry over too.
Priority:	High
Frequency of Use:	Once per matching round.
Flow of Events:	1. The Admin navigates to the "Matching Round" dashboard. 2. The System displays cycle status and matching controls. 3. The Admin selects "Execute Matching". 4. The System verifies Student eligibility, excluding any Student who already holds a match this cycle or who declined or let an offer time out in an earlier round. 5. The System runs student-proposing deferred acceptance over the Students' preference rankings and the Supervisors' applicant rankings, within each listing's open slots. A Student is never matched to a listing whose Supervisor left them unranked. 6. If there is a subsequent round and rollover is enabled, the System queues the rollover of unmatched eligible Students' applications. 7. The System commits match results to storage and logs an immutable entry with Admin ID and timestamp. 8. The System displays summary metrics including matched students, unmatched count, and average preference rank.

Alternative Flows:	-
Exceptions:	E1: Matching initiated prior to deadline closure 1. The System rejects execution and informs the Admin that applications or supervisor ranking periods remain active. E2: Process already executed for the current round 1. The System blocks redundant execution to prevent duplicate match generation.
Includes:	-
Special Requirements:	-
Assumptions:	All submitted rankings are final upon deadline lapse.
Notes and Issues:	-

2.1.4 ReviewMatchingResults

Use Case ID:	#1-4		
Use Case Name:	ReviewMatchingResults		
Created By:	Edwin Tan	**Last Updated By:**	Ian Tan
Date Created:	29th August 2026	**Date Last Updated:**	19th October 2026

Actor:	Admin
Description:	Admin verifies supervisor approvals, handles manual reassignments, and publishes finalized matching outcomes to students and supervisors.
Preconditions:	1. The Admin is signed in. 2. The automated matching process has been executed.

	3. Proposed matches exist in the system.
Postconditions:	Approved matches are published, slots are updated, and users can view their matching result for latest matching round
Priority:	High
Frequency of Use:	Once per matching round
Flow of Events:	1. The Admin navigates to the "Matching Round" page. 2. The System displays the proposed matches along with supervisor approval statuses. 3. The Admin confirms that matches have been approved by the respective Project Supervisors. 4. The Admin clicks "Publish Results to Users". 5. The System presents a confirmation dialog. 6. The Admin clicks "Confirm". 7. The System transitions matching results to the published state and triggers automated notifications to students. 8. Students may accept or decline their match offer before the start of the subsequent matching round's reviewing period, after which it automatically transitions to 'Timed out'. 9. Unmatched students or those from subsequent rounds will wait for the next round's execution.
Alternative Flows:	AF-S3: Manual Reassignment of Student 1. The Admin selects "Reassign" next to a match that has not yet been offered to the Student. 2. The System displays available alternative projects with remaining slots. 3. The Admin selects an alternate project listing, enters a justification of at least 10 characters, and confirms. 4. The System decrements available slots on the new project and increments slots on the original project. 5. The System logs the previous allocation, new allocation, initiating Admin, and timestamp to the audit log.

	6. The System refreshes the proposed match list.
Exceptions:	E1: Admin attempts to publish unapproved matches 1. The System prevents publishing and displays pending supervisor approvals requiring resolution.
Includes:	PublishResults
Extends:	None (ManualAssignStudent extends this use case)
Special Requirements:	Reassignment operations must be atomic to prevent slot over-allocation.
Assumptions:	Project Supervisors complete their match approval reviews before publishing.
Notes and Issues:	Manual overrides must be logged before final release.

2.1.5 PublishResults

Use Case ID:	#1-5		
Use Case Name:	PublishResults		
Created By:	Edwin Tan	**Last Updated By:**	Edwin Tan
Date Created:	29th August 2026	**Date Last Updated:**	29th August 2026

Actor:	Admin
Description:	Allows an Administrator to publish the approved matching results so that students and project supervisors can view their final allocations.
Preconditions:	1. The matching process has been executed for the current round. 2. Project supervisors have completed their approvals for the proposed matches.

Postconditions:	Matching outcomes are transitioned to the published state, and automated notifications are sent to affected users.
Priority:	High
Frequency of Use:	High (at the conclusion of each matching round)
Flow of Events:	1. The Admin navigates to the "Matching Round" page. 2. The Admin confirms that matches have been approved by the respective Project Supervisors. 3. The Admin clicks "Publish Results to Users". 4. The System transitions matching results to the published state. 5. The System triggers automated notifications to students and project supervisors.
Alternative Flows:	AF-1: Admin cancels the publication request prior to final confirmation: 1. Return to the review dashboard.
Exceptions:	E1: Admin attempts to publish results before all required supervisor approvals are completed. 1. The System rejects the publish action and displays an error warning regarding pending approvals. 2. The System remains on the review page.
Includes:	-
Extends:	-
Special Requirements:	Results must only be publishable if approved by the relevant Project Supervisor
Assumptions:	The matching round data is fully recorded and ready for release.
Notes and Issues:	-

2.1.6 ManualAssignStudent

Use Case ID:	#1-6		
Use Case Name:	ManualAssignStudent		
Created By:	Nicholas Ng	**Last Updated By:**	Ian Tan
Date Created:	29th August 2026	**Date Last Updated:**	19th October 2026

Actor:	Admin
Description:	Admin manually assigns a Student to a project. This function shall be reserved for when exceptional circumstances mandate a manual override of URECA Project assignments.
Preconditions:	The Admin is signed in.
Postconditions:	Students are manually assigned to a URECA Project.
Priority:	High
Frequency of Use:	Low (A few times per Application Cycle)
Flow of Events:	1. Admin reviews the matching round results. 2. Admin clicks on “Manual Assign” next to a Student’s entry. 3. Admin selects a published URECA Project with an open slot and enters a justification of at least 10 characters. 4. Admin clicks “Confirm”. 5. The System assigns the Student and records the assignment, justification, initiating Admin and timestamp in the audit log.
Alternative Flows:	- AF-1: Unassign Student. The Admin selects “Unassign” next to an assigned Student and enters a reason of at least 10 characters. The System frees the slot, notifies the Project Supervisor (and the Student, if they had been offered the place), and records the action in the audit log.

Exceptions:	E1: Student is already assigned to a URECA Project 1. The System displays a warning message, and the URECA Project is not assigned. 2. Admin unassigns the current URECA Project from the Student (AF-1). 3. Admin repeats the ManualAssignStudent use case from Step 2.
Includes:	-
Extends:	- ReviewMatchingResults
Special Requirements:	-
Assumptions:	-
Notes and Issues:	-

2.1.7 ManageUserAccounts

Use Case ID:	#1-7		
Use Case Name:	ManageUserAccounts		
Created By:	Edwin Tan	**Last Updated By:**	Ian Tan
Date Created:	29th August 2026	**Date Last Updated:**	19th October 2026

Actor:	Admin
Description:	Admin creates, modifies, or deactivates user accounts in the system
Preconditions:	The Admin is signed in.
Postconditions:	The user's account details, assigned roles, or activation status are updated in persistent storage.
Priority:	High

Frequency of Use:	Occasionally, mostly at the start of a cycle
Flow of Events:	1. The Admin navigates to the "Manage Users" dashboard. 2. The System displays the full directory of users with role and status filters. 3. The Admin selects a specific user account to edit. 4. The Admin updates the user's details or role (Student, Project Supervisor, Admin). 5. The Admin clicks "Save". 6. The System commits the changes and logs the action.
Alternative Flows:	AF-S3: Deactivate User Account: 1. The Admin selects "Deactivate" next to an active user. 2. The System prompts for confirmation. 3. The Admin confirms, and the System marks the user account as deactivated and ends the user's active sessions.
Exceptions:	E1: Creation of Admin role by unauthorized user: 1. The System rejects the role change if the current session does not hold Admin privileges.
Includes:	-
Special Requirements:	Admin account creation and role elevation operations must be strictly restricted to existing authenticated Admins.
Assumptions:	All registered user accounts have valid NTU domain emails.
Notes and Issues:	Deactivated users shall have their active sessions immediately terminated.

2.1.8 MaintainAuditLog

Use Case ID:	#1-8		
Use Case Name:	MaintainAuditLog		
Created By:	Edwin Tan	**Last Updated By:**	Ian Tan
Date Created:	29th August 2026	**Date Last Updated:**	19th October 2026

Actor:	Admin
Description:	Admin views immutable audit records of all administrative operations: cycle and round changes, matching runs, manual assignments and reassignments, publication of results, and account changes.
Preconditions:	The Admin is signed in.
Postconditions:	Audit log entries are retrieved and rendered in read-only mode
Priority:	Medium
Frequency of Use:	After matching cycles or as needed for review
Flow of Events:	1. The Admin navigates to the "Audit & Overrides" tab. 2. The System displays timestamped records of all administrative operations along with initiating Admin IDs. 3. The Admin filters or inspects specific log entries.
Alternative Flows:	-
Exceptions:	E1: Modification attempt: 1. The System strictly prevents any edit or delete operations on audit records.
Includes:	-

Special Requirements:	Audit log entries must be write-once, append-only, and completely immutable to all users including Admins.
Assumptions:	Every administrative operation listed in REQ-AUDIT-1 automatically triggers an audit log write.
Notes and Issues:	Log entries must persist across cycle resets for compliance and tracking.

2.2 For Functional Requirement #2 (Student)

2.2.1 QueryUrecaListings

Use Case ID:	#2-1		
Use Case Name:	QueryUrecaListings		
Created By:	Nicholas Ng	**Last Updated By:**	Ian Tan
Date Created:	26th August 2026	**Date Last Updated:**	19th October 2026

Actor:	Student
Description:	Student browses and queries available URECA Project Listings using keyword search and/or filters.
Preconditions:	1. The Student is signed in. 2. A pre-application period OR an active matching round is ongoing.
Postconditions:	-
Priority:	High
Frequency of Use:	Very High
Flow of Events:	1. The Student selects “View Available Projects” 2. The System displays the first page of available URECA Project Listings in a table view, with certain key information like title, offering organization, and Project Supervisor displayed.

	3. The Student enters a keyword. 4. The Student clicks the “Search” button. 5. The System returns listings that match the query.
Alternative Flows:	AF-S3: The Student uses additional filters/search criteria to query the listings. 1. The Student chooses to filter by school, Project Supervisor name, or topic tag. 2. The System returns listings that match all search/filter criteria. AF-S6: The Student chooses to return to an unfiltered view of the listings. 1. The Student clears all search criteria and clicks “Search”. 2. The System returns to the default and unfiltered paginated view of all URECA Projects Listings.
Exceptions:	E1: The search request from the server times out. 1. The System informs the Student accordingly via an error message.
Includes:	-
Special Requirements:	Each query request shall complete within 5 seconds.
Assumptions:	There will be ample URECA Project Listings posted by Project Supervisors during an active Application Cycle.
Notes and Issues:	-

2.2.2 SubmitUrecaApplication

Use Case ID:	#2-2		
Use Case Name:	SubmitUrecaApplication		
Created By:	Nicholas Ng	**Last Updated By:**	Ian Tan
Date Created:	26th August 2026	**Date Last Updated:**	19th October 2026

Actor:	Student

Description:	Student applies for a URECA Project by submitting an Application to the desired URECA Project Listing
Preconditions:	1. Students are signed in. 2. The current round's Application Window is open. 3. At least one URECA Project Listing is available for application. 4. Student has completed their profile.
Postconditions:	An Application for the desired URECA Project has been submitted to a Project Supervisor for consideration.
Priority:	High
Frequency of Use:	Medium (a few times per Student)
Flow of Events:	1. Students view the full details of a URECA Project Listing. 2. Students click "Apply". 3. Students write a Statement of Interest of at least 50 characters and may attach their uploaded supporting documents. 4. Students rank the project in order of preference (RankAppliedProject). 5. Students click "Submit". 6. The System validates the submitted inputs and documents. 7. The System submits the application for Project Supervisor consideration.
Alternative Flows:	AF-S3: Student clicks “Back to Project” 1. The System displays a confirmation prompt about unsaved Application data. 2. Student clicks “Confirm”. 3. The System returns to displaying the full descriptive view of the selected URECA Project.

Exceptions:	E1: Student leaves the Statement of Interest blank or under 50 characters. 1. The System displays the error message to the Student upon clicking “Submit”, and remains on the Application page. E2: Student uploads documents that are not in PDF form. 1. The System rejects the upload while displaying the appropriate error feedback to the Student. E3: The round's Application Window ends while Students are editing their application before submission. 1. The Student clicks “Submit” 2. The System rejects the submission and displays the reason for rejection. 3. The System returns to displaying the full details of the selected URECA Project, while greying and disabling interactivity of the “Apply” button.
Includes:	RankAppliedProject
Special Requirements:	The Statement of Interest must be at least 50 characters long. The System rejects an application if the Student's profile is incomplete, the Student is ineligible, the listing is not published or has no open slots, or the Student already has an active application to that listing (REQ-APPLY-3).
Assumptions:	-
Notes and Issues:	-

2.2.3 RankAppliedProject

Use Case ID:	#2-3		
Use Case Name:	RankAppliedProject		
Created By:	Edwin	**Last Updated By:**	Ian Tan
Date Created:	26th August 2026	**Date Last Updated:**	19th October 2026

Actor:	Student

Description:	This feature allows a student to arrange and order their submitted URECA project applications in order of preference during the active application window.
Preconditions:	1. The Student has submitted at least one URECA project application. 2. The current round's Application Window is open.
Postconditions:	The student's priority rankings for their submitted applications are successfully recorded and saved in the system.
Priority:	High
Frequency of Use:	Medium (a few times per Student)
Flow of Events:	1. The Student navigates to their submitted applications dashboard. 2. The System displays the list of currently submitted URECA project applications. 3. The Student assigns or adjusts the preference numbers for each submitted project. 4. The Student clicks "Save Rankings". 5. The System validates that all active applications have unique, sequential preference numbers assigned. 6. The System records and saves the updated ranking order.
Alternative Flows:	AF-1: Student chooses to clear and reset their current preference ordering before saving. 1. The Student clicks "Clear Rankings", and the System resets all preference numbers to unranked. 2. The Student enters new rankings and proceeds back to the main flow at Step 4. AF-2: Student withdraws an application that is still under review. The Student clicks "Withdraw" and confirms; the

	System withdraws the application and renumbers the remaining preference ranks.
Exceptions:	E1: Student attempts to submit ranking changes outside of the active application window. 1. The System displays an error message upon clicking "Save", notifying the Student that the window is closed, and prevents the update. E2: Student assigns duplicate or non-sequential ranking numbers to the applications. 1. The System rejects the submission upon clicking "Save" and displays an error message indicating that rankings must be unique. 2. The System remains on the ranking page for the Student to correct the sequence.
Includes:	-
Special Requirements:	Rankings must be unique for each active application submitted. Must be strictly inaccessible outside of the active application window.
Assumptions:	The student has active project applications to rank.
Notes and Issues:	-

2.2.4 SelfProposeUrecaApplication

Use Case ID:	#2-4		
Use Case Name:	SelfProposeUrecaApplication		
Created By:	Nicholas Ng	**Last Updated By:**	Ian Tan
Date Created:	26th August 2026	**Date Last Updated:**	19th October 2026

Actor:	Student
Description:	Students self-propose a URECA Project and submit it to their desired Project Supervisor for approval.
Preconditions:	1. Students are signed in. 2. The current round's Application Window is open.
Postconditions:	A self-proposed URECA Application has been submitted for Project Supervisor consideration.
Priority:	High
Frequency of Use:	Low (Most Students will apply to listed URECA Projects)
Flow of Events:	1. Student clicks on “Propose a Project”. 2. The System displays the Application page for a Self-Proposed URECA Project. 3. Students fill up their proposed URECA Project title, writeup, research interests, supporting documents, and choice of Project Supervisor. 4. Students click “Submit”. 5. The System validates all required fields. 6. The System submits the Application for Project Supervisor consideration.
Alternative Flows:	AF-S3: Student clicks on the “Browse Projects” tab without submitting the Application. 1. The System displays a confirmation prompt about unsaved Application data. 2. Student clicks “Confirm”. 3. The System returns to displaying the URECA Project Listings in table view.
Exceptions:	E1: Student leaves one or more required field(s) blank. 1. The System displays the error message to the Student upon clicking “Submit”, and remains on the Application page. E2: Student uploads documents that are not in PDF form. 2. The System rejects the upload while displaying the appropriate error feedback to the Student. E3: The Application Window ends while Students are editing their application before submission.

	1. The Student clicks “Submit” 2. The System rejects the submission and displays the reason for rejection. 3. The System remains on the proposal page, while greying and disabling interactivity of the “Submit” button.
Includes:	-
Special Requirements:	The project description is limited to 500 words and the statement to 300 words, and at least one research interest is required (REQ-PROPOSE-1).
Assumptions:	-
Notes and Issues:	-

2.2.5 ManageMatchingOutcome

Use Case ID:	#2-5		
Use Case Name:	ManageMatchingOutcome		
Created By:	Edwin Tan	**Last Updated By:**	Ian Tan
Date Created:	29th August 2026	**Date Last Updated:**	19th October 2026

Actor:	Student
Description:	Students review the published matching allocation and decide whether to accept or decline the offered project slot.
Preconditions:	1. The Student is signed in. 2. Matching results have been published by an Admin.
Postconditions:	Match status is recorded as accepted or declined; accepting automatically closes out all other active applications
Priority:	High

Frequency of Use:	Once per cycle.
Flow of Events:	1. The Student navigates to "My Applications". 2. The System displays the "Matching Round Complete" banner and project match offer. 3. The Student reviews the project title and assigned Project Supervisor. 4. The Student clicks "Accept Match". 5. The System displays a warning modal stating that accepting will withdraw all other pending applications. 6. The Student selects "Confirm". 7. The System records the acceptance and automatically sets all other applications submitted by the Student to withdraw.
Alternative Flows:	AF-S4: Student declines match offer 1. The Student clicks "Decline". 2. The System displays a confirmation dialog warning that the student will be ineligible for subsequent matching rounds. 3. The Student clicks "Confirm". 4. The System marks the allocation status as declined. 5. The System flags the Student account as ineligible for any subsequent matching rounds in the current cycle. AF-S5: Student takes no action before the start of the subsequent matching round's reviewing period 1. The System automatically transitions the match status to 'Timed out' 2. The System flags the Student account as ineligible for subsequent rounds.
Exceptions:	E1: Student attempts to access outcomes before Admin publication 1. The System shows current application submission status and hides the offer acceptance interface.

Includes:	ViewMatchingResult
Special Requirements:	Immediate in-app visual confirmation upon selection.
Assumptions:	Each student receives at most one match offer per cycle.
Notes and Issues:	Once confirmed, acceptance cannot be undone through the student UI.

2.2.6 ManageProfile

Use Case ID:	#2-6		
Use Case Name:	ManageProfile		
Created By:	Edwin Tan	**Last Updated By:**	Edwin Tan
Date Created:	29th August 2026	**Date Last Updated:**	29th August 2026

Actor:	Student
Description:	Student completes, views, and updates personal academic details, completed modules, research interests, and transcript documents.
Preconditions:	The Student is signed in.
Postconditions:	Updated profile data and uploaded transcript are saved to the student record.
Priority:	High
Frequency of Use:	Upon first sign-in and occasionally updated.

Flow of Events:	1. The Student navigates to "My Profile". 2. The System displays the student's name, CGPA, year of study, completed modules, and fields of research interest. 3. The Student updates fields and uploads an updated academic transcript PDF. 4. The Student clicks "Save changes". 5. The System validates the file and inputs, then saves the profile.
Alternative Flows:	-
Exceptions:	E1: Invalid transcript format or file size > 5 MB: 1. The System rejects the upload and displays an error message.
Includes:	UploadTranscript
Special Requirements:	File upload must be restricted to PDF format with a maximum size limit of 5 MB
Assumptions:	First-time sign-in requires completion of the profile before browsing or applying for projects
Notes and Issues:	CGPA and module data may optionally be extracted from the transcript in future releases

2.3 For Functional Requirement #3 (Project Supervisor)

2.3.1 ManageProjectDetails

Use Case ID:	#3-1		
Use Case Name:	ManageProjectDetails		
Created By:	Robin Teoh	**Last Updated By:**	Ian Tan

Date Created:	26th August 2026	**Date Last Updated:**	19th October 2026

Actor:	Project Supervisor
Description:	Allows a Project Supervisor to create, update, and close research project listings.
Preconditions:	1. Project Supervisor is logged in and authenticated. 2. Listings may be created before an Application Cycle opens or during it.
Postconditions:	The project listing is created, updated, or closed, and its current state is saved in the system.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. The system displays the Project Supervisor's existing project listings and prompts them to select CREATE, UPDATE, or CLOSE. 2. If the Project Supervisor selects CREATE, the system prompts for the project title, description, and prerequisites. 3. The Project Supervisor enters the number of available slots. 4. The Project Supervisor adds topic tags to the project. 5. The Project Supervisor selects PUBLISH or SAVE AS DRAFT. 6. The system validates the entered data and saves the listing as published, or as a draft.
Alternative Flows:	AF-S2: If the Project Supervisor selects UPDATE 1. The system displays the selected project's existing details for editing. 2. The Project Supervisor modifies the details and selects SAVE. 3. The system validates and saves the updated listing.

	4. If the application closing date has passed, the system rejects the update and displays a message that listings can no longer be modified. AF-S2: If the Project Supervisor selects CLOSE 1. The system prompts the Project Supervisor to confirm closing the project. 2. On confirmation, the system marks the project listing as closed and removes it from student browsing. AF-S6: If the Project Supervisor selects CANCEL 1. The system discards the entered data and returns to the project listing screen.
Exceptions:	E1**:** A required field is missing or invalid. 1. The system rejects the submission. 2. The system displays an error message identifying the invalid field and prompts the Project Supervisor to correct it. 3. The system remains on the project configuration page. E2: The Project Supervisor attempts to reduce the number of slots below the number of applicants already matched to the project. 1. The system rejects the change. 2. The system displays an error message stating the minimum permitted number of slots. 3. The system remains on the project configuration page.
Includes:	-
Special Requirements:	The system must validate all user input data.
Assumptions:	The current matching cycle and its deadlines have been configured by an Administrator.

Notes and Issues:	None

2.3.2 ReviewAndRankStudentApplications

Use Case ID:	#3-2		
Use Case Name:	ReviewAndRankStudentApplications		
Created By:	Robin Teoh	**Last Updated By:**	Ian Tan
Date Created:	26th August 2026	**Date Last Updated:**	19th October 2026

Actor:	Project Supervisor
Description:	Allows a Project Supervisor to review students who applied to their project and rank them in order of preference.
Preconditions:	1. Project Supervisor is logged in and authenticated. 2. The Project Supervisor has at least one published project listing. 3. At least one student has applied to the project. 4. The round's Application Window or Review Window is open.
Postconditions:	The Project Supervisor's applicant rankings are submitted and recorded for use in the matching process.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. The Project Supervisor selects a project to review. 2. The system displays all applicants and a curated list of the top 10 recommended applicants based on supervisor filters (school, minimum CGPA and whether prerequisites are met). 3. The Project Supervisor selects an applicant to review. 4. The system displays the applicant's profile and transcript.

	5. The Project Supervisor returns to the applicant list and arranges the applicants in order of preference. 6. The Project Supervisor selects SUBMIT RANKINGS. 7. The system saves the rankings and confirms submission.
Alternative Flows:	AF-S6: If the Project Supervisor modifies rankings after an earlier submission 1. The system replaces the previously submitted rankings with the new order, provided the ranking-submission deadline has not passed. AF-S7: If the Project Supervisor selects CANCEL 1. The system discards the unsaved ranking order and returns to the applicant list.
Exceptions:	E1: The Project Supervisor attempts to submit rankings after the round's Review Window has closed. 1. The system rejects the submission. 2. The system displays a message stating that the ranking-submission deadline has closed. 3. The system remains on the applicant list page. E2: No Student has applied to the selected project. 1. The system displays a message stating that there are no applicants to rank. 2. The system disables the ranking and submission functions for that project.
Includes:	-
Special Requirements:	Applicant transcripts must be viewable within the system without requiring external software. Applicants left unranked are not matched to that listing.
Assumptions:	Students have completed their profiles before applying.

Notes and Issues:	Rankings must be submitted before the deadline set by the Administrator; supervisors who do not submit rankings may have their applicants matched on student preference alone.

2.3.3 ApproveMatchingResults

Use Case ID:	#3-3		
Use Case Name:	ApproveMatchingResults		
Created By:	Robin Teoh	**Last Updated By:**	Ian Tan
Date Created:	26th August 2026	**Date Last Updated:**	19th October 2026

Actor:	Project Supervisor
Description:	Allows a Project Supervisor to review students proposed to be matched to their project and approve or reject each proposed match.
Preconditions:	1. Project Supervisor is logged in and authenticated. 2. The Administrator has executed the matching process. 3. At least one proposed match exists for the Project Supervisor's project.
Postconditions:	Each proposed match is recorded as approved or rejected. Approved matches become eligible for publication by the Administrator.
Priority:	High
Frequency of Use:	Low
Flow of Events:	1. The system displays the students proposed to be matched to the Project Supervisor's project. 2. The Project Supervisor selects a proposed match to review. 3. The system displays the student's profile and application details. 4. The Project Supervisor selects APPROVE.

	5. The system records the match as approved and marks it as eligible for publication. 6. Steps 2 to 5 are repeated until all proposed matches have been processed.
Alternative Flows:	AF-S5: If the Project Supervisor selects REJECT 1. The system prompts the Project Supervisor to confirm the rejection. 2. On confirmation, the system records the match as rejected. 3. The system records the rejection reason for Administrator review and frees the slot. 4. Return to step 3. AF-S5: If the Project Supervisor selects CANCEL 1. The system returns to the list of proposed matches without recording a decision.
Exceptions:	E1: The Project Supervisor attempts to approve or reject a proposed match after the matching results have been published. 1. The system prevents the approval or rejection. 2. The system displays a message stating that the matching results are final. 3. The system displays the proposed matches in a read-only state.
Includes:	-
Special Requirements:	None
Assumptions:	The Administrator will review recorded rejections and reassign affected students before publishing results.
Notes and Issues:	None

2.3.4 ManageSelf-ProposedProjects

Use Case ID:	#3-4		
Use Case Name:	ManageSelf-ProposedProjects		
Created By:	Edwin Tan	**Last Updated By:**	Ian Tan
Date Created:	29th August 2026	**Date Last Updated:**	19th October 2026

Actor:	Project Supervisor
Description:	Allows a Project Supervisor to review research proposals pitched directly by students and either accept & register or decline them.
Preconditions:	1. The Project Supervisor is logged in. 2. At least one student has submitted a self-proposed project to this supervisor.
Postconditions:	The proposal status is updated, and the student is either registered to the supervisor or notified of rejection.
Priority:	High
Frequency of Use:	Several times during an active application cycle
Flow of Events:	1. The Project Supervisor navigates to the "Proposals from students" tab. 2. The System displays all proposals submitted by students directed to the supervisor. 3. The Project Supervisor reviews the proposal title, writeup, research tags, and supporting documents. 4. The Project Supervisor clicks "Accept & register". 5. The System registers the student to a new one-slot project under the supervisor, withdraws the student's other open applications, makes the student ineligible for later rounds, and notifies the student.
Alternative Flows:	AF-S4: Supervisor declines proposal:

	1. The Project Supervisor clicks "Decline" and may add a note. The System marks the proposal as declined and updates the student's dashboard.
Exceptions:	E1: Application Cycle has closed: 1. The System blocks acceptance and displays a notice. The proposal can still be declined.
Includes:	-
Special Requirements:	Accepting a proposal must automatically trigger an email/system notification to the student.
Assumptions:	A student can only be registered to one confirmed URECA project per cycle.
Notes and Issues:	Direct supervisor acceptance bypasses the automated matching algorithm round.

2.4 For Functional Requirement #4 (User)

2.4.1 ViewNotifications

Use Case ID:	#4-1		
Use Case Name:	ViewNotifications		
Created By:	Edwin Tan	**Last Updated By:**	Edwin Tan
Date Created:	29th August 2026	**Date Last Updated:**	29th August 2026

Actor:	User (Student, Project Supervisor, Administrator)
Description:	Allows an authenticated user to view incoming and historical system notifications regarding status changes, application updates, cycle dates, and matching outcomes.
Preconditions:	The User is logged in and authenticated.

Postconditions:	Notifications are displayed, and selected unread items are updated to a read state in persistent storage. .
Priority:	Medium
Frequency of Use:	High (frequently checked during active application and matching periods)
Flow of Events:	1. The User clicks on the "Notifications" tab in the navigation menu. 2. The System queries and displays the list of relevant timestamped notifications (e.g., application status changes, new applicant alerts, match results, or deadline updates). 3. The User selects a notification to view its complete message details. 4. The System marks the selected notification as read and updates the unread badge counter.
Alternative Flows:	AF-S3: User navigates via notification interactive link The User clicks an embedded link inside a notification (e.g., "View Application" or "View Match Result"). The System redirects the User directly to the target record page
Exceptions:	E1: Notification service fetch timeout or server failure 1. The System displays an error message informing the user that notifications could not be retrieved. 2. The System provides a "Retry" action.
Includes:	-
Special Requirements:	In-app notifications must be delivered and displayed within 1 minute of the triggering event
Assumptions:	All relevant status-change triggers automatically persist notification records for affected user accounts.
Notes and Issues:	Email notifications, where applicable, are dispatched asynchronously within 15 minutes of the event.

2.4.2 SignIn

Use Case ID:	#4-2		
Use Case Name:	SignIn		
Created By:	Ian Tan	**Last Updated By:**	Ian Tan
Date Created:	31th August 2026	**Date Last Updated:**	19th October 2026

Actor:	User (Student, Project Supervisor, Administrator)
Description:	Allows a registered, verified user to securely access the URECAMatch platform using their email address and password.
Preconditions:	The User has a registered and activated account.
Postconditions:	The user is authenticated and granted access to their respective dashboard.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The User navigates to the sign-in page. 2. The System displays input fields for NTU email address and password. 3. The User enters their credentials and clicks "Sign in". 4. The System validates the credentials against the user database. 5. If valid, the System authenticates the user and redirects them to their respective dashboard.
Alternative Flows:	AF-1: User clicks "Forgot password?" - The System redirects the user to the password reset flow.
Exceptions:	

	E1: Invalid credentials provided. 1. The System displays an error message and prevents access. E2: Account is deactivated. 1. The System prevents sign-in and displays an error message. E3: Account is not yet verified. The System prevents sign-in and prompts the User to complete e-mail verification.
Includes:	-
Special Requirements:	Password input must be masked, with an option to reveal it. Session shall expire after 30 minutes of inactivity.
Assumptions:	Users have already successfully completed the registration and email verification process.
Notes and Issues:	-

2.4.3 RegisterAccount

Use Case ID:	#4-3		
Use Case Name:	RegisterAccount		
Created By:	Ian Tan	**Last Updated By:**	Ian Tan
Date Created:	31th August 2026	**Date Last Updated:**	19th October 2026

Actor:	User (Student or Project Supervisor; Admin accounts are created by an Admin)
Description:	Allows a user to create a new account using an NTU email address and password, followed by an email verification process to activate the account.
Preconditions:	None (User is not yet registered).

Postconditions:	A new user account is created in the system and activated upon successful verification.
Priority:	High
Frequency of Use:	High (at start of application cycle)
Flow of Events:	1. The User navigates to the registration page. 2. The System displays input fields for full name, NTU email address, password, and account role selection (Student or Project Supervisor). 3. The User enters details and clicks "Create account". 4. The System validates the email domain and checks for existing accounts. 5. The System sends a 6-digit verification code, valid for 10 minutes, to the provided email address. 6. The User enters the verification code. 7. The System verifies the code and activates the account.
Alternative Flows:	AF-1: User requests a new verification code if the previous one expired or was not received.
Exceptions:	E1: Invalid NTU email domain provided. E2: Email address already associated with an existing account. E3: Incorrect or expired verification code entered. E4: Password shorter than 8 characters. The System rejects the registration and displays the password rule.
Includes:	-
Special Requirements:	Registration is restricted to NTU email domains (@e.ntu.edu.sg or @ntu.edu.sg) or approved external domains. Password input must be masked.
Assumptions:	Users have access to their provided email account to receive the verification code.
Notes and Issues:	-